



ROOT CAUSE UNLOCKED

# The Complete Patient's Guide to *Hip* Pain

*A Root Cause Approach to Understanding and Recovering from Hip Pain*

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## BEFORE YOU START

**Disclaimer:** *This ebook is for educational purposes only and does not constitute medical advice. Always consult with a qualified healthcare provider before starting any new treatment, supplement, exercise program, or dietary change. Do not start, stop, or change any prescribed medication without speaking to the prescriber. This guide does not diagnose your condition. If you experience the warning signs described in Chapter 11, seek medical evaluation promptly.*

**A note on the evidence in this guide.** *Every numbered citation points to a specific published study that was checked against the source paper before it was used here. Where the research on a topic is strong, this guide says so. Where the research is thinner than clinical experience suggests, or where a claim rests mainly on years of hands on practice rather than a body of trials, this guide says that too, plainly, rather than dressing clinical judgment up as settled science.*

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# 1. Understanding Hip Pain

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If your hip has been bothering you, you have probably already noticed that it does not behave the way you expected. Maybe it aches deep in the groin when you stand up from a chair. Maybe it is a sharp, nagging pain on the outside of your hip that flares up the moment you lie on that side at night. Maybe it started as low back pain and slowly migrated into your buttock and hip. Hip pain is not one condition. It is a region, and a great many different structures in that region can become the source of your discomfort.

This guide takes an integrative, root cause approach to hip pain, the same approach I use in my practice at Root Health in Garner, North Carolina. That approach starts from a simple premise: the place where you feel pain is not always the place where the problem began. A tight, weak, or poorly coordinated muscle group around the hip and pelvis can quietly overload a tendon, a joint, or a nerve until it finally speaks up as pain. Understanding those relationships, rather than chasing the painful spot alone, is what this guide is built around.

You will also notice this guide spends real time on gluteus medius and gluteus minimus tendinopathy, often still called greater trochanteric pain syndrome or, incorrectly, hip bursitis. In my clinical experience, these two muscles are involved in the majority of hip presentations I see, whether the pain is on the outside of the hip itself or in a related pattern elsewhere in the leg or low back. That observation lines up closely with what the research literature now says about lateral hip pain, which is covered in detail in Chapter 4.

Hip pain is also, encouragingly, very often manageable with active, structured care rather than a quick procedure. Most of the conditions in this guide respond to a combination of load management, targeted strengthening, and attention to the muscles and movement patterns around the hip and pelvis, not to rest and waiting. That is the direction this guide points you in throughout.

One more thing before you continue. This guide organizes your thinking about hip pain. It does not diagnose you. A qualified clinician, examining your specific hip, history, and movement pattern, is the only way to know for certain what is driving your pain. Use this guide to understand your options and to ask better questions at your next appointment, not as a substitute for that appointment.

## 2. Anatomy of the Hip: The Structures at Play

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The hip joint itself is a deep, stable ball and socket joint, the femoral head seated inside the acetabulum of the pelvis. That depth and stability is part of why the hip rarely dislocates the way a shoulder does. It is also part of why hip pain is so often generated not by the joint surfaces themselves but by the dense network of muscles, tendons, and ligaments that surround and move that joint.

### The Bony and Joint Structures

The femur, the pelvis, and the sacrum form the bony architecture of the hip and pelvic girdle. Where the sacrum meets the pelvis on either side, you have the sacroiliac, or SI, joints, small joints that transmit load between your spine and your legs. A cartilage rim called the labrum lines the acetabulum and deepens the socket, and a layer of articular cartilage covers the joint surfaces themselves.

### The Muscle Groups I Focus On

Rather than list every muscle that crosses the hip, it is more useful to understand the functional groups I evaluate in nearly every hip patient, because these are the groups most often out of balance.

**The hip flexors: psoas major and iliacus.** Together often called the iliopsoas, these muscles run from the lumbar spine and the inside of the pelvis down to the top of the femur, and their job is to flex the hip and help stabilize the lumbar spine and pelvis. In my clinical experience, a tight or poorly functioning psoas and iliacus is a frequent contributor to hip dysfunction, particularly in people who sit for long stretches of the day. It is worth being direct about the evidence here: the idea of a “tight psoas” is one of the most widely repeated concepts in manual and movement based practice, and it has real clinical currency, but it is not backed by a deep body of high quality trial evidence specifically linking psoas tightness to hip pain. This guide treats it as clinical reasoning, not an established fact, and says so again in Chapter 6.

**The external rotators of the hip.** A small group of deep muscles, including the piriformis, that rotate the thigh outward and help stabilize the femoral head in the socket during walking and standing. When these muscles are chronically short or overactive, my clinical experience is that they can alter hip mechanics and contribute to lateral and posterior hip discomfort. As with the psoas, this is a clinical pattern more than a heavily researched one, and it is presented that way in Chapter 6.

**The tensor fasciae latae, or TFL.** A muscle at the front of the outer hip that blends into the iliotibial band running down the outside of the thigh. It is frequently implicated in lateral hip and knee complaints in clinical practice. A 2020 systematic review and meta-analysis in *Sports Medicine* examined whether the TFL actually differs in structure or activation between people with lower limb musculoskeletal conditions, including lateral hip pain, and people without. The reviewers found only weak, inconsistent evidence for the common clinical assumptions about the TFL’s role, and concluded that those assumptions are not well investigated and are poorly supported by current research.<sup>1</sup> That does not mean the TFL is irrelevant. It means the confident

claims sometimes made about it outrun what has actually been studied, which is worth knowing before anyone recommends an aggressive TFL-focused protocol to you.

**The gluteus medius and gluteus minimus.** These two muscles sit on the outer surface of the pelvis, beneath the larger gluteus maximus, and their primary job is to stabilize your pelvis every time you take a step on one leg. My clinical experience is that these two muscles are involved in almost every hip case I evaluate, and as you will see in Chapter 4, that observation is strongly supported by the research literature on lateral hip pain specifically.

## How These Groups Work Together

None of these muscle groups act alone. The hip flexors, external rotators, TFL, and glute medius and minimus all cross the same joint and share the job of controlling your pelvis during standing, walking, and single leg tasks like climbing stairs. When one group is tight, weak, or slow to activate, the others are forced to compensate, and that compensation is frequently where pain begins. This interconnected view of the hip carries through the rest of this guide, and it is the foundation of the regional interdependence model discussed fully in Chapter 6.

### 3. Common Causes of Hip Pain

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Hip pain is generally grouped by where you feel it: the front of the hip and groin, the outside of the hip, or the back of the hip and buttock. This is a useful starting framework, because different structures tend to produce pain in different locations, although there is real overlap.

**Anterior, or front of hip and groin, pain** is often related to the hip joint itself. In younger, more active adults this can include labral tears or femoroacetabular impingement, a mechanical conflict between the femur and the socket. In older adults, hip osteoarthritis is a common driver of anterior groin pain. Front of hip pain can also be referred from structures inside the abdomen or pelvis, which is one more reason self-diagnosis is not a good idea.

**Lateral, or outside of hip, pain** is what most people mean when they describe a nagging ache over the bony point of the hip that worsens with lying on that side, walking, or climbing stairs. This is the territory of greater trochanteric pain syndrome, covered in full in Chapter 4, and it is by far the most common presentation my approach is built around.

**Posterior, or back of hip and buttock, pain** can come from the hip joint itself, from deep gluteal structures including the piriformis and the sciatic nerve as it passes beneath it, or it can be referred pain traveling down from the lumbar spine. A 2021 review in *American Family Physician* laid out this same anterior, lateral, and posterior framework for evaluating adult hip pain, and specifically noted that posterior hip pain includes referred pain from lumbar spinal pathology as well as deep gluteal syndrome with sciatic nerve entrapment.<sup>2</sup> That overlap between the low back and the hip is a recurring theme in this guide and is central to Chapter 6.

#### A Common Misdiagnosis Worth Naming Early

For decades, persistent pain over the outer hip was routinely labeled “trochanteric bursitis,” implying inflammation of the small fluid filled sac, the bursa, that cushions the tendons near the greater trochanter. Current evidence tells a different story, and it is important enough that the next chapter is devoted to it entirely.

#### A Brief Note on the Sacroiliac Joint

Some readers of this guide will have pain that is genuinely coming from the sacroiliac joint rather than the hip itself. SI joint pain typically sits low and to the side of the spine, over the dimples at the top of the buttock, and it is a real and commonly under-recognized contributor to what people describe as low back and hip pain, affecting an estimated 15 to 25 percent of people with axial low back pain according to a widely cited review in *Anesthesia & Analgesia*.<sup>3</sup> Because SI joint dysfunction is experienced by most people as chronic low back and buttock pain rather than as hip pain specifically, the full discussion of how it develops and how it is managed, including the role of sacroiliac support belts, lives in my companion guide, *The Complete Patient's Guide to Low Back Pain*. If your pain sits more over your low back and buttock than over the side of your hip,

or if it has never resolved despite treatment aimed at your back, that guide and a proper evaluation for SI joint involvement are worth pursuing alongside this one.



## 4. The Real Story Behind Lateral Hip Pain: Gluteal Tendinopathy

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If you have lateral hip pain, the single most important thing this guide can tell you is this: it is very likely not bursitis.

For years, the working assumption in medicine was that pain over the outer hip, especially pain that worsened with lying on that side, was caused by inflammation of the trochanteric bursa, a small cushioning sac near the top of the femur. Research over the past two decades has substantially overturned that assumption. A landmark study published in *Arthritis and Rheumatism* examined 24 patients with the classic clinical picture of greater trochanteric pain syndrome using MRI. Gluteus medius tendon pathology, either tearing or tendinitis, was found in the large majority of patients, while true bursal distension was uncommon and, notably, did not occur in the absence of gluteus medius pathology.<sup>4</sup> In other words, when the bursa was inflamed at all, the tendon was already involved too. The bursa was a bystander, not the primary problem.

A comprehensive 2015 review in *Sports Medicine* confirmed and extended this picture, stating plainly that tendinopathy of the gluteus medius and gluteus minimus tendons is now recognized as the primary local source of lateral hip pain.<sup>5</sup> This condition is most common in midlife, occurs in both athletes and people who do not regularly exercise, and affects women more than men. It interferes with sleep, particularly side lying, and with common weight bearing tasks, which is exactly the pattern many of my patients describe: pain that wakes them at night and pain that builds with walking or stairs.

This is precisely the pattern my clinical model points to directly. I have long observed that the gluteus medius and gluteus minimus are almost always involved in hip presentations, and the research on greater trochanteric pain syndrome now bears that observation out with real specificity. What was once dismissed as a vague, catch all “bursitis” diagnosis has a much more precise and much more treatable explanation: overloaded, irritated hip abductor tendons.

### Why This Distinction Actually Matters to You

Calling this condition bursitis matters because it changes how people try to treat it. Bursitis suggests inflammation that might respond to rest and anti-inflammatory measures alone. Tendinopathy is a different problem, one driven by how the tendon is being loaded over time, and it generally responds best to a structured, progressive loading program rather than rest. Chapter 7 walks through what that looks like in practice.

### The Role of Compression

One of the more clinically useful findings from the tendinopathy research is the role of compression. The 2015 *Sports Medicine* review found that the combination of excessive compressive load and high tensile load

on the gluteal tendons is thought to be the most damaging combination, and that joint position, particularly excessive hip adduction, where the thigh crosses toward or past the midline of the body, is a key factor in this condition.<sup>5</sup> Every time your femur moves into an adducted position, whether from standing with your weight shifted onto one hip, sitting with your legs crossed, or sleeping curled on your side without support between your knees, the iliotibial band and the gluteal tendons are compressed against the bony point of the hip.

This has a direct and somewhat counterintuitive practical implication, addressed fully in Chapter 7: some of the stretches people instinctively reach for to relieve lateral hip pain, particularly stretches that pull the leg across the body into adduction, can actually provoke the very compression that is aggravating the tendon in the first place. Stretching alone, and the wrong kind of stretching in particular, is not a reliable fix for this condition.

## What About Injections?

Corticosteroid injections are commonly offered for greater trochanteric pain syndrome, and they can and do provide real short term relief for many people. The strongest evidence available on how injections compare with a structured exercise program comes from a large, well conducted trial known as the LEAP trial, published in *BMJ* in 2018. Researchers randomly assigned 204 adults with confirmed gluteal tendinopathy to one of three groups: a physiotherapy led program of education plus exercise over eight weeks, a single corticosteroid injection, or a wait and see approach with no active treatment.<sup>6</sup>

The results were notable. At eight weeks, both the education plus exercise group and the corticosteroid injection group reported significantly better global improvement and less pain than the wait and see group, which is exactly what you would expect: doing something beats doing nothing. But the education plus exercise group did not just match the injection group, it outperformed it, both on the global rating of improvement and on reported pain scores at eight weeks. At the one year follow up, the education plus exercise group continued to show better global outcomes than the corticosteroid injection group, while the two groups' pain scores had converged somewhat by that point, with both still ahead of the wait and see group.<sup>6</sup>

The honest, plainly stated takeaway from this trial is not that injections do nothing. They clearly helped compared with doing nothing at all. The takeaway is that a structured program of education and exercise did at least as well as an injection in the short term and did better on the measure of overall improvement a year later. An injection is not a solution on its own, and it should not be presented to you as one. If you are offered a corticosteroid injection for lateral hip pain, it is entirely reasonable to ask your provider whether a loading program is being paired with it, because the best evidence available says that combination, or exercise alone, is where the durable improvement comes from.

## 5. How Hip Pain Is Diagnosed

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A thorough evaluation for hip pain typically combines your history, a physical examination, and, when needed, imaging.

### History and Physical Examination

Your provider will want to know where exactly your pain is located, what makes it better or worse, whether it changes with specific positions like lying on your side or crossing your legs, and whether you have any of the warning signs covered in Chapter 11. Physical examination typically includes assessing your hip's range of motion, your strength in hip abduction and other directions, your gait, and specific provocative tests. One classic finding worth knowing about is the Trendelenburg sign, in which the pelvis drops on the side opposite the affected hip when you stand on the affected leg alone, a sign of gluteus medius weakness or dysfunction. In the original MRI study of greater trochanteric pain syndrome discussed in Chapter 4, this sign was the most accurate of the physical tests examined for predicting an actual gluteus medius tendon tear, correctly identifying about 73 percent of tears and correctly ruling out the condition in about 77 percent of people who did not have a tear.<sup>4</sup>

### Imaging: When It Helps

Not everyone with hip pain needs imaging right away. A 2021 review in *American Family Physician* on evaluating hip pain in adults notes that plain radiography of the hip and pelvis is usually the appropriate initial imaging test, with ultrasound or MRI reserved for cases where the diagnosis remains unclear or where findings would change management, such as suspected labral tears, femoroacetabular impingement, or gluteus medius tendon tears, which often have good outcomes with surgery when appropriate and where earlier imaging and referral can matter.<sup>2</sup> For most cases of lateral hip pain consistent with gluteal tendinopathy, a clinical diagnosis based on history and examination is often sufficient to begin a structured treatment program, without needing advanced imaging first.

### Differential Diagnosis: Why Location Alone Is Not Enough

As Chapter 3 introduced, hip pain located at the front, side, or back of the hip tends to point toward different structures, but there is meaningful overlap, and referred pain from the lumbar spine can mimic pain originating in the hip itself.<sup>2</sup> This is exactly why this guide does not attempt to diagnose you, and why a proper clinical evaluation, one that also assesses your lumbar spine and pelvis rather than only the hip in isolation, is genuinely necessary before starting a treatment plan.

## 6. Regional Interdependence: Why the Hip Rarely Works Alone

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This chapter is the connective tissue of my clinical model, and it deserves to be read as its own idea rather than a footnote to the others.

The formal term for this way of thinking is regional interdependence, and it has a real foothold in the physical therapy and rehabilitation literature. A widely cited paper on the model, published in the *Journal of Manual and Manipulative Therapy*, describes the underlying premise directly: seemingly unrelated impairments in remote anatomical regions of the body may contribute to and be associated with a patient's primary report of symptoms, and interventions directed at one region of the body will often have effects at remote and seemingly unrelated areas.<sup>7</sup> In plainer language, the place that hurts and the place that is driving the problem are frequently not the same place.

This is the lens I apply to hip dysfunction. A tight or weak psoas and iliacus, tight external rotators, and an overactive TFL are not simply local nuisances. They change how your femur sits in the socket and how much work your gluteus medius and minimus have to do to stabilize your pelvis with every step. When those upstream muscles are not doing their share, the tendons at the outer hip end up absorbing load they were not built to handle alone, which loops directly back to the gluteal tendinopathy story in Chapter 4.

Some supporting research for this interconnected view comes from outside the hip specifically. A 2025 study in *Scientific Reports* examined hamstring strength and hip mobility, including hip extension and hip internal rotation range of motion, in young adults with and without clinically defined lumbar instability, and found that both hamstring strength and hip mobility were strongly associated with pain and disability, independent of each other.<sup>8</sup> This is not a study of hip pain patients specifically, and its authors were careful to describe their findings as hypothesis generating rather than diagnostic, but it is a genuine, measured example of hip function and low back symptoms moving together, which is exactly the kind of relationship the regional interdependence model predicts.

### Functional Leg Length Inequality

The fifth pillar of my clinical model is functional leg length inequality, a difference in effective leg length that arises from muscular, postural, or pelvic factors rather than an actual difference in bone length. In my clinical experience, this kind of asymmetry can contribute not only to hip pain but to low back, knee, ankle, and foot pain as well, because every step is being taken on an uneven foundation.

This is genuinely an area where, as with the psoas, the trial evidence is thinner than the clinical pattern would suggest, and it is worth saying so directly rather than overselling it. A systematic review published in *Archives of Physical Medicine and Rehabilitation* examined shoe lift interventions for leg length discrepancy in adults with painful musculoskeletal conditions and found that leg length discrepancy was indeed associated with

low back pain, scoliosis, and osteoarthritis of the hip and knee in the literature, but that the overall quality of the included studies was very low to poor, and that only one included study was a randomized controlled trial.<sup>9</sup> Within that limited evidence, the results were still worth noting: in the studies that reported it, the large majority of people treated with a shoe lift experienced partial or complete pain relief, and the one randomized trial found meaningful pain reduction in the treated group.<sup>9</sup> A more recent, narrower systematic review in the *Journal of the American Podiatric Medical Association* looked specifically at foot orthoses and low back pain in people with leg length inequality and found that every included study reported a significant reduction in low back pain with orthosis use, although the evidence on whether orthoses actually changed walking mechanics was inconsistent.<sup>10</sup> A separate review connecting foot and ankle mechanics to chronic low back pain likewise noted that excessive pronation has been linked to leg length discrepancies that contribute to pelvic tilt and low back pain, describing the connection as part of a broader kinetic chain running from the foot to the low back.<sup>11</sup>

Taken together, this is a real, plausible, and clinically observed pattern that does not yet have the kind of large, rigorous trial evidence behind it that the gluteal tendinopathy research does. My use of functional leg length assessment and correction, where appropriate, reflects clinical judgment built on this pattern and on years of hands on evaluation, not on a settled body of high quality research. That distinction matters, and this guide is not going to claim more certainty than the evidence supports.

## **The Practical Upshot**

Regional interdependence is the reason a thorough hip evaluation looks beyond the hip itself, checking your lumbar spine, your pelvis, your leg length, and the muscle groups covered in Chapter 2. It is also the reason that a treatment plan built only around the painful spot, without addressing what is feeding into it, so often stalls out. The exercise and movement approach in the next chapter is built with this whole picture in mind.



# 7. Movement and Exercise for Hip Recovery

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**The most important message in this chapter: load management and progressive strengthening have the strongest evidence behind them of anything discussed in this guide, and they should be the center of your recovery plan, not an afterthought.**

## Understanding Load, Not Just Pain

A useful way to think about tendinopathy, including gluteal tendinopathy, comes from a foundational 2009 paper in the *British Journal of Sports Medicine*, which proposed that tendon pathology exists on a continuum rather than as a single fixed injury.<sup>12</sup> Depending on how a tendon has been loaded over time, it can move along that continuum toward reactive irritation or toward more degenerative change, and different points on that continuum call for different treatment intensity. The practical takeaway for you is straightforward: your tendon's pain is telling you something about how it has been loaded, not simply that it is damaged, and the right amount and right kind of loading, not none at all, is usually the way forward.

## The Evidence Centerpiece: Education Plus Exercise

As discussed fully in Chapter 4, the LEAP trial found that an eight week, physiotherapy led program combining education about load management with a progressive exercise program outperformed a single corticosteroid injection, both in the short term and at one year follow up, for people with confirmed gluteal tendinopathy.<sup>6</sup> That program is the single best supported intervention for this condition among everything covered in this guide, and it should be the anchor of any recovery plan for lateral hip pain.

## What to Avoid: Compressive Stretches

Because excessive hip adduction compresses the gluteal tendons against the greater trochanter, and because that compression is implicated as a driver of gluteal tendinopathy, some commonly recommended hip stretches can do more harm than good if you have this condition.<sup>5</sup> Be cautious with any stretch that pulls your leg across the midline of your body, such as pulling your knee toward your opposite shoulder while lying on your back, or a standing IT band stretch where you cross one leg behind the other and lean away. Sitting with your legs crossed for long periods and standing with your weight sunk into one hip, a posture sometimes called hip hitching, create the same compressive load and are worth minimizing during a flare. This does not mean all stretching is off limits, but it does mean stretching should be selected thoughtfully rather than reached for automatically, ideally with guidance from a clinician familiar with this condition.

## Gentle Isometrics to Begin

For an irritable tendon, isometric holds, contracting the muscle without moving the joint, are often a reasonable and well tolerated starting point before progressing to more dynamic strengthening.

**Isometric Hip Abduction Against a Wall:** 1. Stand with the outside of your affected hip a few inches from a wall 2. Press your leg outward into the wall without letting your body move 3. Hold for 20 to 30 seconds, breathing normally 4. Repeat 3 to 5 times

## Progressive Strengthening

Once the tendon tolerates isometric loading comfortably, progressive strengthening of the hip abductors, particularly the gluteus medius and gluteus minimus, becomes the priority.

**Sidelying Hip Abduction:** 1. Lie on your unaffected side with both knees slightly bent 2. Keeping your hips stacked and your top hip from rolling backward, lift your top leg toward the ceiling 3. Avoid lifting the leg too high, which shifts the work away from the target muscles 4. Perform 2 to 3 sets of 10 to 15 reps, progressing to a resistance band around the thighs as tolerated

**Standing Hip Hike:** 1. Stand on your affected leg on a small step, letting the other foot hang off the edge 2. Slowly lower the hanging hip toward the floor, then use your standing hip's outer muscles to lift it back level 3. Perform 2 to 3 sets of 10 reps

**Bridge with Band:** 1. Lie on your back, knees bent, a resistance band around your thighs just above the knees 2. Press your knees gently outward against the band while lifting your hips into a bridge 3. Hold briefly at the top, lower slowly 4. Perform 2 to 3 sets of 10 to 15 reps

## Addressing the Whole Chain

Consistent with the regional interdependence model in Chapter 6, a complete program also addresses hip flexor mobility and general hip and pelvic control, not the outer hip alone.

**Half Kneeling Hip Flexor Stretch, Modified:** 1. Kneel on one knee with the other foot forward in a lunge position 2. Tuck your pelvis under and gently shift your weight forward, feeling a stretch through the front of the kneeling hip 3. Keep the front knee tracking over the foot, and avoid letting the kneeling side hip drift outward into adduction 4. Hold for 20 to 30 seconds, repeat 2 to 3 times per side

**Bird Dog:** 1. Start on hands and knees 2. Extend one arm forward and the opposite leg straight back, keeping your pelvis level 3. Hold for 5 to 10 seconds, return to start 4. Perform 2 to 3 sets of 8 reps per side

## Walking

Regular walking remains one of the simplest and most useful tools for most people with hip pain, as it loads the hip abductors rhythmically and predictably. If walking aggravates your specific pain pattern, shortening your stride slightly and avoiding uneven or heavily cambered surfaces can help, and this is a good topic to bring to your provider if walking consistently flares your symptoms.

## **A Word on Pacing**

Progress in tendon rehabilitation is rarely linear. A tendon that has been irritated for months did not become irritated overnight, and it will not resolve overnight either. Expect some day to day variation, and treat a single bad day as information rather than a sign that the whole plan has failed.



# 8. Integrative and Physical Therapies

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## 8.1 Physical Therapy

A skilled physical therapist can build and progress the kind of graded loading program described in Chapter 7, assess the movement patterns and muscle imbalances discussed in Chapter 6, and adjust your program as your tendon's tolerance changes. Given the strength of the evidence behind structured education and exercise for gluteal tendinopathy specifically, physical therapy is a reasonable first line recommendation for most people with lateral hip pain.<sup>6</sup>

## 8.2 Chiropractic Care

Chiropractic evaluation and care can address joint mobility in the hip, pelvis, and lumbar spine, and can incorporate the regional interdependence assessment described in Chapter 6, including screening for functional leg length asymmetry and pelvic mechanics that may be contributing to your hip pain. This guide does not have a hip specific randomized trial of chiropractic care to cite here, and it is more honest to say so than to borrow evidence from a different condition. What chiropractic care can reasonably offer is a whole person evaluation of the kinetic chain feeding into your hip, paired with hands on treatment and a home exercise program.

## 8.3 Corticosteroid Injections

As detailed in Chapter 4, corticosteroid injections can provide genuine short term relief for gluteal tendinopathy, but the best available randomized evidence found that an eight week education and exercise program matched or outperformed a single injection at every time point measured, including one year later.<sup>6</sup> An injection may be a reasonable part of a plan, particularly to reduce pain enough to begin exercise more comfortably, but it functions best as a bridge to active rehabilitation rather than as a standalone treatment.

## 8.4 Shockwave Therapy

Extracorporeal shockwave therapy, which delivers acoustic pulses to irritated tendon tissue, is used clinically for gluteal tendinopathy that has not responded to a structured exercise program. It is a reasonable option to discuss with your provider if first line care has stalled, though it is generally considered a second line option after a genuine trial of load management and exercise rather than a replacement for it.

## 8.5 Heat and Cold

Heat and cold are simple, low risk, and can offer symptomatic comfort. Cold applied for 15 to 20 minutes may help calm an acutely irritable tendon after a flare, while heat can ease general stiffness and muscle tension in

the surrounding hip flexors and external rotators. Neither replaces a loading program, but either can make daily life more comfortable while you work through one.

## **8.6 Massage and Soft Tissue Work**

Massage and soft tissue work targeting the hip flexors, external rotators, and TFL can provide short term relief of tension and may support your ability to tolerate the exercise program in Chapter 7. As with chiropractic care, this guide does not have hip specific trial evidence to cite for massage, and presents it here as a reasonable, low risk adjunct rather than a primary treatment.

## 9. Nutrition for Joint and Tendon Health

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What you eat will not repair a tendon on its own, and this guide is not going to claim otherwise. A general pattern that supports connective tissue repair and manages systemic inflammation is nonetheless a sensible part of a whole person recovery plan.

### **Foods to Emphasize**

Favor foods that supply the building blocks of tendon and cartilage repair alongside a broadly anti-inflammatory pattern of eating: quality protein sources such as wild caught fish, pasture raised poultry, eggs, and legumes; colorful vegetables and fruits, particularly leafy greens and berries; omega-3 rich foods like salmon, sardines, walnuts, and flaxseed; and healthy fats such as olive oil and avocado. Bone broth and collagen containing foods supply amino acids used in connective tissue repair, though as with any supplement or food based approach, expect this to support your overall recovery environment rather than to resolve a specific tendon problem on its own.

### **Foods to Minimize**

Refined carbohydrates and added sugars, deep fried foods, heavily processed meats, and excessive alcohol are worth minimizing, both for general inflammatory tone and because they tend to displace more nutrient dense choices from your plate.

### **Hydration**

Adequate hydration supports tissue health broadly. There is no need to chase a specific number. Drink regularly through the day and let thirst and pale colored urine guide you, unless your provider has given you specific fluid guidance for a heart or kidney condition.

### **Weight and Joint Load**

Carrying additional body weight increases the mechanical load passing through your hips with every step, which matters for both hip osteoarthritis and tendon based conditions like gluteal tendinopathy. Reaching and maintaining a healthy weight is worth pursuing for hip health among many other reasons, though it should be approached as one part of a broader plan rather than a precondition for starting the exercise program in Chapter 7.

# 10. Lifestyle Modifications: Sleep, Weight, and Footwear

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## 10.1 Sleep Positioning

Because gluteal tendinopathy is aggravated by compression from hip adduction, how you sleep matters directly. If you sleep on your side, place a pillow between your knees to keep your top hip from adducting and compressing the irritated tendon against the mattress, and consider sleeping with a pillow supporting the top leg more broadly if lying on the affected side is particularly painful. Some people find temporary relief sleeping on their back with a pillow under their knees, or on the unaffected side.

## 10.2 Standing and Sitting Posture

Avoid standing for long periods with your weight shifted into one hip, the hip hitching posture mentioned in Chapter 7, which compresses the outer hip tendons on the weight bearing side. When sitting, avoid prolonged leg crossing for the same reason, and take regular breaks to change position rather than holding any one posture for hours.

## 10.3 Footwear and Leg Length

If a functional leg length asymmetry has been identified as part of your evaluation, as discussed in Chapter 6, your provider may recommend a heel lift or orthotic on one side. The evidence here, again, is real but modest: a systematic review of shoe lifts for leg length discrepancy found low quality but broadly positive evidence for pain relief, alongside a clear call for higher quality research to determine which patients benefit most and by how much correction.<sup>9</sup> This is a reasonable intervention to try under professional guidance, with realistic expectations set from the outset. Supportive, well fitted footwear generally, with adequate cushioning for daily walking, is a reasonable and low risk addition regardless of leg length findings.

## 10.4 Activity Modification During a Flare

During a significant flare, temporarily reducing high compressive or high impact activities, such as running on cambered roads, deep squatting past 90 degrees, or prolonged standing on one leg, can calm symptoms while you continue the lower irritation isometric and strengthening work from Chapter 7. This is a temporary adjustment, not a long term restriction, and the goal is always to return to full activity through graded loading rather than through indefinite avoidance.

# 11. The In-Clinic Pre-Treatment Protocol

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Understand what this is for, because the distinction matters.

None of this rebuilds tissue. The loading and strengthening work in the previous chapters does that. What this protocol does is reduce the muscle tone, stiffness and tissue sensitivity that make the strength work harder to tolerate and less pleasant to do, particularly in the early weeks when things are at their most irritable.

It is preparation, not treatment. I use it in the clinic before the working part of a session, and I give it to patients to use at home before theirs. Patients who do it tend to tolerate the loading better and stick with it longer, and sticking with it is the actual mechanism by which recovery happens.

If you have to choose between doing this and doing your strength work, do your strength work.

## **Percussion device (Hypervolt), round or flat tip.**

- Tensor fasciae latae and hip flexors, 1 to 2 minutes.
- All three glutes, 1 to 2 minutes per side.
- Adductors, gently, 1 minute per side.

## **Vibrating sphere (Hypersphere Mini) or roller (Vyper or foam roller).**

- Glutes, 2 to 3 minutes per side. Lie on your back and move the leg up and down, or sit and roll.
- Hip flexor, psoas and iliacus, 1 to 2 minutes per side.
- IT band, side lying, from hip to just above the knee, 1 to 2 minutes per side.

## **Safety, and I mean these.**

- Avoid all of this over an acutely injured area in the first 48 to 72 hours.
- Reduce pressure over anything actively inflamed.
- Skip percussion therapy entirely if you have a pacemaker or another implanted electronic device.
- Avoid aggressive pressure over varicose veins or compromised skin.
- Start light. Pain must not exceed 6 out of 10. Back off immediately for sharp or shooting pain.
- Stop and seek advice if you get numbness, tingling, or weakness.

## 12. Red Flags: When to Seek Immediate Care

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Most hip pain is not an emergency, and the great majority of what is covered in this guide responds to the kind of structured, active care described in earlier chapters. A small number of presentations, however, warrant prompt medical evaluation, and it is worth knowing them.

### **Seek prompt medical evaluation if you experience:**

- Hip pain following a fall or direct trauma, particularly if you are older or have known osteoporosis risk, because this raises concern for a hip fracture
- Fever accompanying joint pain, which can indicate infection
- Unexplained weight loss along with hip or pelvic pain
- Night pain that is not relieved by any change in position, which is different from pain that simply worsens with lying on the affected side
- Progressive weakness, numbness, or other neurological symptoms in the leg
- Groin pain severe enough that you are unable to bear weight on the leg at all

These situations call for prompt evaluation, not panic, and most people who experience one of them do not turn out to have a serious underlying condition. But they are exactly the situations where this guide, or any self-directed exercise program, should be set aside in favor of a timely appointment or, for a suspected fracture or infection, urgent care.

# 13. Building Your Personal Recovery Plan

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Use this as a starting framework to discuss with your provider, not as a fixed prescription. Your specific findings, particularly which structures are involved and how irritable your symptoms currently are, should shape how quickly you move through these stages.

## **Weeks 1 to 2: Calm and Assess**

Begin with a proper evaluation that looks at your hip, your lumbar spine, your pelvis, and your movement pattern as a whole, consistent with the regional interdependence approach in Chapter 6. Start gentle isometric hip abduction work if tolerated, adjust your sleep position to avoid compressive adduction, and minimize hip hitching and prolonged leg crossing. Avoid aggressive stretching into hip adduction during this period.

## **Weeks 3 to 6: Build Tolerance**

Progress from isometrics into the sidelying abduction, standing hip hike, and banded bridge work described in Chapter 7, adding hip flexor mobility and general core and pelvic control work like bird dog. Continue addressing sleep positioning and standing posture. If pain is significant, this is a reasonable point to discuss a corticosteroid injection with your provider as a bridge to more comfortable exercise participation, understanding from the LEAP trial evidence that the injection works best paired with, not instead of, the exercise program.<sup>6</sup>

## **Weeks 7 to 12: Strengthen and Restore**

Progress resistance in your hip abductor strengthening, reintroduce walking distance and pace gradually, and begin addressing any identified functional leg length asymmetry with your provider's guidance. This is also a reasonable point to consider shockwave therapy if progress has plateaued despite consistent loading work.

## **Beyond 12 Weeks: Maintenance**

Continue hip abductor and general lower body strength work two to three times per week indefinitely, since deconditioning is a common precursor to recurrence. Maintain attention to sleep positioning and posture as ongoing habits rather than temporary fixes, and revisit your provider if symptoms plateau or recur despite consistent adherence to your program.



# Final Thoughts

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Hip pain, and lateral hip pain in particular, has been misunderstood and mismanaged for a long time under the label of bursitis. The research is now clear that for most people with pain over the outer hip, the real story is a tendon that has been asked to do more than it was prepared for, usually because the muscles and movement patterns feeding into that tendon, the psoas and iliacus, the external rotators, the TFL, and the pelvis and low back beyond, were not sharing the load the way they should have been.

That reframing matters because it points toward a genuinely different and genuinely more hopeful path forward: not rest and injections alone, but understanding your load, retraining your gluteus medius and minimus, and addressing the whole kinetic chain around your hip rather than the painful spot in isolation. That is the root cause approach this guide has tried to walk you through, and it is the approach I use with my patients at Root Health every day.

You do not need to have all the answers before you start. You need an accurate evaluation, a plan that respects both the strong evidence behind load management and the honest limits of what is known about some of the other contributing factors, and the patience to let a tendon or a movement pattern change over weeks rather than days.

If you would like a thorough, root cause evaluation of your hip pain and the patterns that may be feeding into it, I offer a free Root Discovery Session at Root Health. You can schedule yours at [rootcauseunlocked.com/discovery](https://rootcauseunlocked.com/discovery). You can also reach the office at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

# 14. References

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*This guide was compiled from peer-reviewed medical literature and clinical experience. Every citation was verified against the published record. Where a claim reflects clinical judgment rather than trial evidence, this guide has said so directly. It is intended to give you knowledge and options to discuss with your healthcare provider. It is not a substitute for individual medical care, and it does not diagnose your condition or promise a cure.*

**This document is a DRAFT pending Dr. Seay's clinical review.**

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